

Document 6. Elasticities — PED, YED, XED, PES

Topic overview. Elasticity measures the responsiveness of one variable to a change in another. Four elasticities are tested in 9570: Price Elasticity of Demand (PED), Income Elasticity of Demand (YED), Cross Elasticity of Demand (XED) and Price Elasticity of Supply (PES).

Section A. Important Definitions

Term	Definition
PED	$\frac{\% \Delta Q_d}{\% \Delta P}$ for the same good. Always negative; we use the absolute value.
YED	$\frac{\% \Delta Q_d}{\% \Delta Y}$ where Y is income. Sign indicates good type.
XED	$\frac{\% \Delta Q_d^A}{\% \Delta P^B}$ where A and B are different goods. Sign indicates relationship.
PES	$\frac{\% \Delta Q_s}{\% \Delta P}$. Always non-negative.
Total Revenue (TR)	$P \times Q$. Used with PED to predict revenue effects of price changes.

Section B. Interpretation Bands

Elasticity	Value	Interpretation
PED	$ PED > 1$	Price elastic. Cut price $\rightarrow$ TR rises.
PED	$ PED < 1$	Price inelastic. Cut price $\rightarrow$ TR falls.
PED	$ PED = 1$	Unit elastic. TR unchanged.
PED	$ PED = \infty$	Perfectly elastic (horizontal DD).
PED	$ PED = 0$	Perfectly inelastic (vertical DD).
YED	$YED > 0$	Normal good. If $YED > 1$ luxury; if $0 < YED < 1$ necessity.
YED	$YED < 0$	Inferior good.
XED	$XED > 0$	Substitutes. The larger, the closer.
XED	$XED < 0$	Complements. The larger in magnitude, the stronger.
XED	$XED = 0$	Unrelated goods.
PES	$PES > 1$	Supply elastic.
PES	$PES < 1$	Supply inelastic.

Section C. Determinants

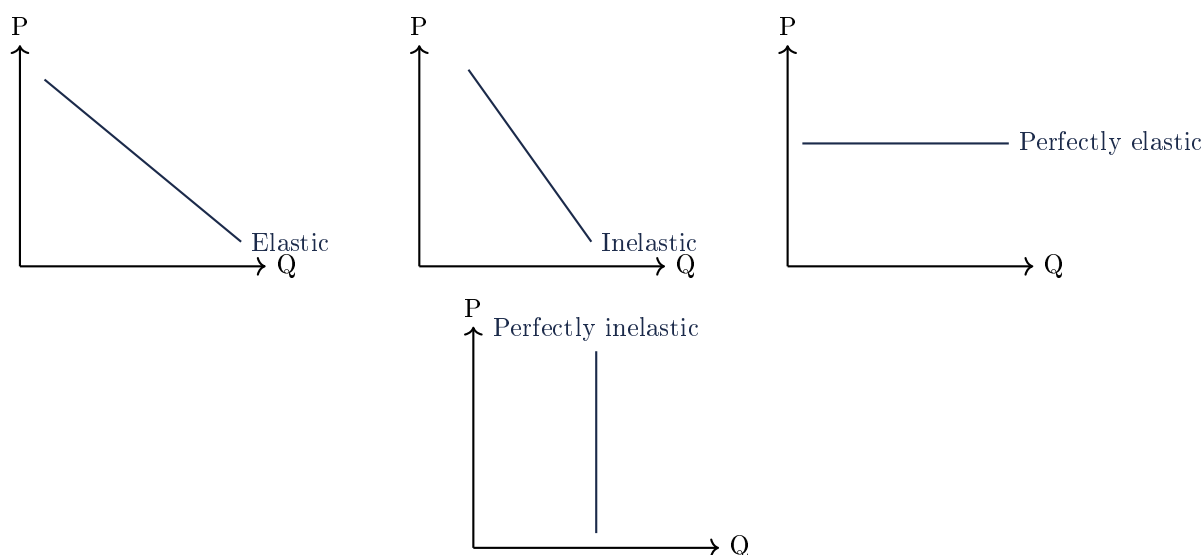
PED & PES determinants (SNIT): Substitutes / Necessities, Income proportion spent on the good, Time horizon.

Additional PES determinants: spare capacity, ease of storage, mobility of factors of production, time horizon.

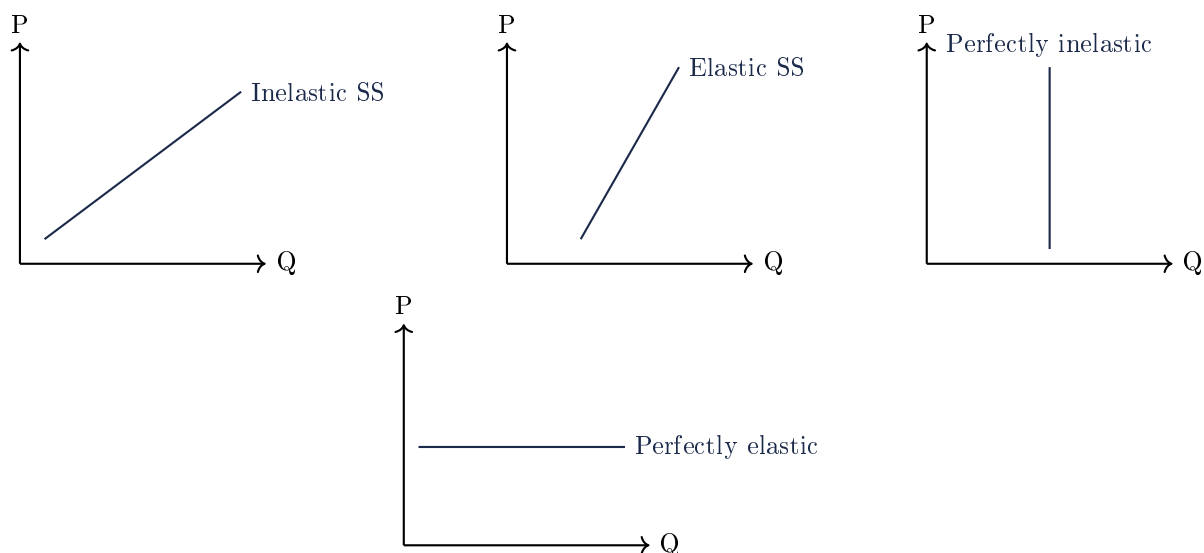
YED determinants (NL): Nature of the good (necessity, luxury, inferior), Level of income of the consumer.

XED determinants: closeness of the substitute / complement relationship.

Section D. Diagrams — PED Variants



Section D1. Diagrams — PES Variants



Section E. Question Type 1: Compute and interpret PED

Technique: substitute into $PED = \frac{\% \Delta Q_d}{\% \Delta P}$; state the band; conclude on revenue effect.

Step 1. State the formula.

Step 2. Compute the percentage changes using endpoint values from the data.

Step 3. Substitute and obtain the elasticity value.

Step 4. Classify into the interpretation band (elastic / inelastic / unit / perfectly).

Step 5. Predict the effect on TR using the inverse rule.

Section F. Question Type 2: Apply elasticity to firm pricing

- Step 1.** Identify whether DD for the firm's product is elastic or inelastic, using SNIT.
Step 2. If elastic, recommend price cuts to raise TR; if inelastic, recommend price increases.
Step 3. Acknowledge limits: PED varies along a linear DD curve; competitor reactions may shift DD; cost considerations matter for profit.
Step 4. Conclude with a firm-specific recommendation.

Section G. Question Type 3: Apply elasticity to government policy

- Step 1.** Identify the policy goal (raise revenue / reduce consumption / stabilise prices).
Step 2. For revenue-raising taxes (e.g. tobacco, fuel) target goods with **inelastic** DD — small fall in Q , large tax revenue.
Step 3. For consumption-reduction (carbon tax) target goods with **elastic** DD or pair the tax with substitutes (public transport).
Step 4. Acknowledge equity (regressive impact on low-income households) and supply-side responses.

Section H. Worked Examples

Worked Example 1: Computing PED

Q. Quantity demanded for a good falls from 100 to 80 units when price rises from \$10 to \$12. Compute PED.

$\% \Delta Q_d = -20\%$. $\% \Delta P = +20\%$. $PED = -20/20 = -1$. **Unit elastic**; TR unchanged.

Worked Example 2: Necessity good and tax

Q. Should the government impose an excise tax on rice?

Rice has inelastic DD (necessity, no close substitute, small budget share). A tax raises revenue with limited consumption fall, but is *regressive* and creates inflationary cost-of-living concerns. Singapore therefore zero-rates rice for GST.

Worked Example 3: XED and product strategy

Q. Coffee shops compete with tea shops. XED of coffee w.r.t. tea = +0.8. Interpret.

Positive but < 1 : substitutes, but not strong. A 10% rise in tea price raises coffee Q_d by 8%. Coffee shops can capture demand from a tea price hike, but should not assume one-for-one substitution.